

ME 610: Advanced Thermodynamics

Course Code	ME 610
Course Name	Advanced Thermodynamics
L-T-P-C	3-1-0-4
Prerequisites	Thermodynamics
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Laws of Thermodynamics**
The first law for open and closed system; steady & transient processes; work and heat transfer; second Law of Thermodynamics for open and closed systems; Local Thermodynamic Equilibrium (LTE) Model, entropy maximum and energy minimum principles. [Lectures]
- **Entropy**
Concept of reversibility; change in entropy in various thermodynamic processes, entropy balance for closed and open systems, mechanism of entropy generation. [Lectures]
- **Single and Multiphase systems**
Maxwell relations; Clausius-Clapeyron equation; Gibbs-Duhem Relation, phase diagrams, corresponding states; phase transition; types of equilibrium and stability; multi- component and multi-phase systems, equations of state. [Lectures]
- **Chemically Reacting System**
Chemical reactions, irreversible reactions, combustion, chemical energy of fuels. [Lectures]
- **Power Generation**
Irreversibilities in a power plant; advanced steam-turbine power plants; advanced gas-turbine power plants, combined steam turbine and gas turbine plants. [Lectures]
- **Refrigeration**
Joule-Thomson expansion, Liquefaction, refrigerator models with heat transfer irreversibilities. [Lectures]
- **Entropy Generation Minimization**
heat transfer, trade-off between competing irreversibilities, principle of thermodynamic isolation, structure of heat exchanger irreversibility, energy storage systems, sensible and latent heat storage. [Lectures]
- **Kinetic theory of gases**
Introduction, basic assumption, molecular flux, equation of state for an ideal gas, collisions with a moving wall, principle of equipartition of energy, classical theory of specific heat capacity. Transport phenomena- intermolecular forces, The Vander Weals equation of state, collision cross section, mean free path. [Lectures]

Text Books

1. Adrian Bejan, **Advance Engineering Thermodynamics**, Wiley, 2006.
2. M.J.Moran and H.N.Shapiro, **Fundamentals Of Engineering Thermodynamics**, John Wiley and Sons.

References

1. F.W. Sears and G.L.Salinger, **Thermodynamics, Kinetic Theory and Statistical Thermodynamics**, 3rd Edition, Narosa Publishing House, New Delhi, 1998.